Buying the Pullback in the Cleanest Trends: An EMA Band Strategy for US Equities, 2007–2026

[Full Name][1], [Full Name][2], [Full Name][3]

[1][Faculty], [University], [City], Indonesia
[2][Faculty], [University], [City], Indonesia
[3][Faculty], [University], [City], Indonesia

Team [Team Name] · Arena: United States · Benchmark: S&P 500 (SPY) · January 2007 to November 2026

ABSTRACT

We study a long-only US equity strategy that ranks stocks by the quality of their trend, buys the leaders when the price pulls back into the band between the 20-day exponential moving averages of opens and closes, and sells after 63 trading days. A concentrated five-position version earns 39.1% a year over its July 2017 to September 2026 development sample. Under the QISC constraints (ten positions of at most 10% of capital, a $2 million liquidity floor and 10 basis points per side) and over January 2007 to November 2026, which adds a decade not used in development, the strategy earns 12.4% a year against 10.8% for the S&P 500, with a lower Sharpe ratio (0.54 against 0.62), a 58% drawdown and a monthly excess return that is not statistically different from zero. It beats the index only in the development period. One component survives every test: within the same month, entries on a band touch beat entries into the same trends without one by 1.1 percentage points over five days and 2.3 points over three months, including in the held-out decade. The pullback entry is a robust timing effect; the portfolio's excess return is not.

Keywords: Momentum; Short-term reversal; Exponential moving average; Technical analysis; Out-of-sample testing; Multiple testing

Submitted to the Quant Investment Strategy Competition (QISC) 2027, Preliminary Round. Every number, table and figure in this paper is produced by `python run.py` in the accompanying code repository (commit `4f9c2e1`).

1. Introduction

Buying a strong stock on a dip is among the most widely practised trading ideas and among the least rigorously tested. It combines two well-documented regularities. Over horizons of months, returns continue: stocks that have risen tend to keep rising (Jegadeesh and Titman, 1993; Moskowitz et al., 2012). Over days to weeks, returns partly reverse (Jegadeesh, 1990; Lehmann, 1990). An investor who selects strong trends but times the purchase to a short pullback should, in principle, earn the first premium at a better price because of the second.

We turn this idea into a fully specified, point-in-time strategy and evaluate it as an investor would have to trade it. Each day we rank US stocks by the quality of their current trend, measured as the trend's return divided by its typical pullback. Among the top-ranked stocks, we buy one when its price dips into the band between the 20-day exponential moving average (EMA) of opens and of closes and then closes back above it, and we sell every position after 63 trading days. The rules were developed on data from July 2017 onward. We test them on the decade before, which played no role in their design, impose the trading constraints of the QISC casebook, and correct for the number of variants we examine.

The evidence separates a robust component from a fragile one. A concentrated five-position version earns 39.1% a year in its development sample but only 13.6% over 2007–2026. The QISC-compliant portfolio earns 12.4% a year against 10.8% for the S&P 500, with more risk and a monthly excess return whose 95% interval includes zero; it beats the index only in the development period and trails it in the held-out decade and in the 2025–2026 out-of-sample window. The timing rule, by contrast, holds everywhere: compared within the same month, entries on a band touch outperform entries into the same top-ranked trends without one by 1.1 percentage points over five days and 2.3 points over three months, and both effects are significant in the held-out decade.

Section 2 sets out the economic rationale and hypotheses, Section 3 the data and sample design, and Section 4 the strategy and evaluation methods. Section 5 reports the main results, Section 6 the robustness tests, Section 7 implementation and risk, and Section 8 concludes.

2. Economic rationale and hypotheses

2.1. *Mechanism*

We classify the edge as behavioural, with a liquidity component. Momentum is commonly attributed to investors who underreact to gradually arriving information (Jegadeesh and Titman, 2001), and the disposition effect, the tendency to sell winners too early, is one channel through which such underreaction arises (Shefrin and Statman, 1985; Frazzini, 2006). Short-term reversal, in contrast, is largely compensation for supplying liquidity to investors who need to trade (Lehmann, 1990; Nagel, 2012). A pullback within a strong trend is where the two effects meet: the trend is intact, but the price has been pushed temporarily below it by selling unrelated to the firm's prospects. The counterparty on the entry day is therefore a holder who sells into the dip, such as a trader taking profits, an investor selling a winner too early, or a trend-follower whose

stop has been triggered. On the exit day, three months later, the typical counterparty is a later momentum buyer.

2.2. *Related evidence*

Technical trading rules have a long and contested record. Moving-average rules appeared profitable on the Dow Jones index (Brock et al., 1992), but much of that evidence weakens once the number of rules searched is accounted for (Sullivan et al., 1999) and once transaction costs and false discoveries are controlled (Bajgrowicz and Scaillet, 2012). At the intraday horizon, none of 7,846 popular rules survives a data-snooping correction in US equities (Marshall et al., 2008). Cross-sectional moving-average signals, which compare stocks with each other rather than time a single index, have held up better (Han et al., 2013). Closest to our entry rule, algorithmically identified Fibonacci retracement zones predict price bounces, although the bounce does not by itself imply a profitable strategy (Tsinaslanidis et al., 2022). Machine-learning studies routinely use moving averages and related indicators as inputs (Patel et al., 2015; Sezer and Ozbayoglu, 2018) but rarely evaluate a complete, cost-aware portfolio. Three forces work against a strategy of this kind: momentum crashes in sharp market rebounds and broad sell-offs (Daniel and Moskowitz, 2016), published anomalies decay (McLean and Pontiff, 2016), and rules selected after inspecting a sample tend to overstate their future performance (Harvey et al., 2016).

2.3. *Hypotheses*

The strategy makes a selection bet and a timing bet, and we test them separately.

H1 (selection). Stocks ranked in the top ten by trend quality earn a positive return in excess of the equal-weighted eligible universe over the following 63 trading days.

H2 (timing). Within the same month, entries into top-ranked stocks on a band touch earn more over the following 5, 21 and 63 trading days than entries into top-ranked stocks without a touch.

H3 (portfolio). A ten-position portfolio that combines both rules earns a positive return in excess of the S&P 500 after costs, in the held-out decade and in the QISC out-of-sample window, not only in the development period.

3. Data and sample design

3.1. *Universe and prices*

The universe starts from every security in the NASDAQ Trader symbol directory (November 2026) listed on NASDAQ, NYSE, NYSE American or NYSE Arca, excluding exchange-traded funds, test issues, warrants, rights, units, preferred shares and notes. Using the country and industry fields of the Nasdaq stock screener, we keep companies located in the United States and drop closed-end funds, which leaves 4,231 names. Daily open, high, low and close prices and volume come from Yahoo Finance, adjusted for splits and dividends, from January 2005 to 11 November 2026. Of the 4,231 names, 3,793 reach a price of at least $1 and a 60-day median traded value of at least $0.5 million at some point; on an average day about 40 pass the trend screen. The benchmark is SPY with dividends reinvested. Zero and negative prices are set to missing, as is any open more than three times above or below the same day's close; no price is forward-filled. We focus on the US

because the screen needs a large universe to find enough clean trends on any given day; the smaller set of liquid IDX stocks, the long-only constraint and auto-rejection limits would leave few candidates.

3.2. *Sample design*

The rules were developed on data from July 2017 to September 2026. We therefore divide the sample into three periods (Table 1). The scores require 200 trading days of history, so the first decisions are made in January 2007.

Table 1. Sample periods.

Period	Dates	Role
Held-out	3 Jan 2007 to 30 Jun 2017	Not used in any design decision; the cleanest out-of-sample test
Development	1 Jul 2017 to 31 Dec 2024	Rules, parameters and filters were chosen on data that include this period
QISC out-of-sample	1 Jan 2025 to 11 Nov 2026	The final 24 months required by the casebook; overlaps the development data up to September 2026

We report the QISC window separately throughout, as the casebook requires, but treat the held-out decade, which includes the 2008 financial crisis, as the more informative out-of-sample test.

3.3. *Survivorship bias*

The symbol directory lists only stocks that trade today, so delisted companies are missing. The bias favours the strategy, because the missing stocks are disproportionately those whose trends ended badly, and it grows with distance from the present. Results for the held-out decade are therefore an upper bound, which strengthens any negative conclusion drawn from them.

4. Strategy and evaluation

4.1. *The EMA band*

For a price series x_t, the exponential moving average with span N is

$$E_t = \alpha x_t + (1-\alpha)E_{t-1} = \sum_{k \geq 0} \alpha(1-\alpha)^k x_{t-k}, \qquad \alpha = \frac{2}{N+1}. \tag{1}$$

With $N = 20$, today's price receives 9.5% of the weight and the weight halves every 6.9 trading days, so E_t approximates the average price paid by the last month's buyers. We compute the EMA on opens (E^O_t) and on closes (E^C_t) and call the space between them the band, with upper edge $U_t = \max(E^O_t, E^C_t)$ and lower edge $L_t = \min(E^O_t, E^C_t)$. A close above the band is above where recent buyers both started and ended the day (Figure 1).

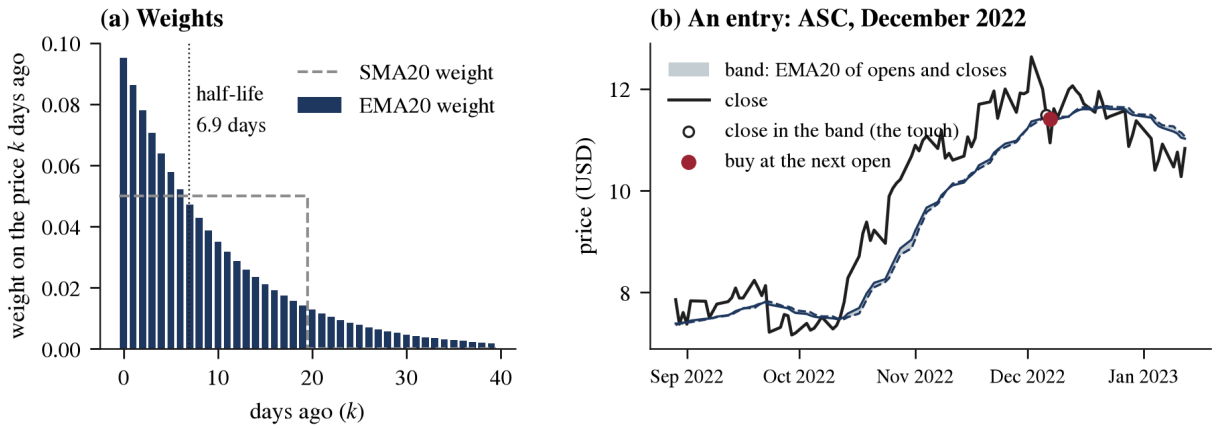


Figure 1. (a) The weight that the EMA20 places on the price k days ago, compared with a 20-day simple moving average. (b) An entry in the baseline portfolio: the close falls into the band, then closes back above it, and the stock is bought at the next open.

4.2. *Trend selection*

A run starts on day $r(t)$, the first day after the latest stretch of more than ten consecutive closes below E^C. A stock passes the screen on day t if its close is above E^C_t and above the 100-day EMA of closes; the run has lasted at least 60 trading days and gained at least 5% in log terms; it contains at least two completed pullbacks of 5% or more, identified on closes with a zigzag filter; no single day contributes more than 20% of the run's log return and no five days more than 55%; and the price and liquidity floors are met. The last conditions exclude runs built on a few jumps rather than steady accumulation. Stocks that pass are ranked by

$$\mathrm{R/R}_t = \frac{C_t / C_{r(t)} - 1}{|\bar{D}_t|}, \tag{2}$$

where C_t is the close and $\bar{D}_t$ the average depth of the completed pullbacks in the current run: the number of its own typical pullbacks the stock has gained. All quantities use prices up to day t, with one exception: a pullback whose low lies before t is counted even if the 5% rebound that confirms it occurs after t. Section 6.4 removes this exception.

4.3. *Entry, exit and position sizing*

A top-ranked stock generates an entry signal when, within the last five trading days, it closed inside the band or less than 1% above it, and today closes above the band:

$$\min_{s \in [t-4,\, t]} \left(\frac{C_s}{U_s} - 1 \right) \le 1\% \quad \text{and} \quad C_t > U_t. \tag{3}$$

Decisions are made at the close of day t and executed at the open of day $t+1$. Each new position receives one tenth of current equity, capped by available cash; positions are not resized while held; every position is sold at the open 63 trading days after entry; and a stock sold in the previous five days cannot be bought again. Uninvested cash earns nothing. Table 2 lists every parameter.

Table 2. Parameters of the baseline portfolio.

Parameter	Baseline portfolio	Definition
Universe	US companies, no funds	Price at least $1; 60-day median traded value at least $2 million
Trend	EMA20 and EMA100	Close above the 20-day and the 100-day EMA of closes
Run	> 10 closes; 60 days; +5%	Closes below EMA20 that end a run; minimum length and log return
Pullbacks	2 of at least 5%	Zigzag reversals on closes within the current run
Gap filter	20% / 55%	Maximum share of the run's log return from the best day / best five days
Ranking	Top 10 by R/R	Eq. (2)
Entry	5 days, 1%	Window and tolerance of the band touch, Eq. (3)
Positions	10 × 10% of equity	Long-only, unlevered; casebook limit of 10% per name
Exit	63 trading days	Sold at the open; no other exit
Costs	10 bp per side	5 bp fees and 5 bp slippage, the casebook minimum

The concentrated variant in Section 5.1 differs in three respects: five positions of 20%, the $0.5 million liquidity screen only, and 5 bp per side.

4.4. *Evaluation*

We report the compound annual growth rate (CAGR), annualised volatility, Sharpe and Sortino ratios, maximum drawdown, the ratio of CAGR to maximum drawdown, and beta and annualised alpha from a regression of monthly returns on SPY. Sharpe ratios are computed from daily returns without subtracting a risk-free rate, for the strategy and the benchmark alike. Confidence intervals for the monthly excess return resample whole months 2,000 times. In the event study, a touch event is the first day of an entry signal for a top-ranked stock and a no-touch event is a day on which a top-ranked stock trends above the band without a recent touch, which corresponds to buying immediately. Returns run from the next open, net of costs, and are also measured in excess of the equal-weighted eligible universe. Because no-touch events cluster in strong markets, we compare the two entries within each month and average the monthly differences. Every variant in Section 6 counts as a trial for the deflated Sharpe ratio of Bailey and López de Prado (2014).

5. Results

5.1. *From the development sample to twenty years*

Table 3 shows how much of the development-sample result survives a longer sample and realistic constraints. The concentrated variant earns 39.1% a year over July 2017 to September 2026, with a monthly excess return of 2.23% whose interval excludes zero. Over 2007–2026, the same rules earn 13.6% a year with a 70% drawdown, most of it in 2008. The QISC-compliant baseline earns 12.4% a year, with a shallower drawdown and the same Sharpe ratio. In both twenty-year columns the excess return over SPY is statistically indistinguishable from zero.

Table 3. The concentrated variant and the baseline portfolio.

Measure	Concentrated, 2017–2026	Concentrated, 2007–2026	Baseline, 2007–2026
Trades	171	361	710
Win rate	56%	52%	54%
Growth per year (SPY)	+39.1% (+15.0%)	+13.6% (+10.8%)	+12.4% (+10.8%)
Sharpe ratio (SPY)	0.99 (0.85)	0.53 (0.62)	0.54 (0.62)
Worst drawdown (SPY)	−47% (−34%)	−70% (−55%)	−58% (−55%)
Monthly return above SPY	+2.23%	+0.65%	+0.29%
95% interval	[+0.18%, +4.51%]	[−0.57%, +1.87%]	[−0.48%, +1.05%]
$1 becomes (SPY)	$20.90 ($3.62)	$12.67 ($7.67)	$10.18 ($7.67)

Note: Concentrated: five positions of 20%, $0.5 million liquidity screen, 5 bp per side. Baseline: Table 2. The first column covers 3 July 2017 to 18 September 2026.

The universe restriction to US companies matters in a concentrated portfolio. Without it, the concentrated variant earns 33.7% rather than 39.1% a year over its development sample, because foreign listings occupy scarce positions and displace or delay some of the largest US winners.

5.2. *The baseline portfolio by period*

Table 4 and Figure 2 report the baseline portfolio over the full sample and in each period. Over nearly twenty years it turns $1 into $10.18 against $7.67 for SPY, with a beta of 0.97 and a volatility of 29.6%, about one and a half times that of the index. In the held-out decade it trails the index (6.3% against 7.5% a year) with a deeper drawdown. In the development period it beats the index (17.8% against 14.4%) but with a lower Sharpe ratio. In the QISC out-of-sample window it trails again (10.7% against 16.1%) with a beta of 1.39. In no period is the excess return distinguishable from zero: H3 is not supported.

Table 4. Performance of the baseline portfolio, January 2007 to November 2026.

Measure	Full sample	Held-out	Development	QISC out-of-sample
Panel A: Return				
Growth per year (CAGR)	+12.4%	+6.3%	+17.8%	+10.7%
S&P 500 growth per year	+10.8%	+7.5%	+14.4%	+16.1%
Monthly return above SPY	+0.29%	−0.01%	+0.48%	−0.00%
95% interval	[−0.48%, +1.05%]	[−0.94%, +0.91%]	[−0.85%, +1.91%]	[−3.72%, +3.85%]
Alpha per year	+3.8%	+1.0%	+4.7%	−5.4%
Panel B: Risk				
Volatility per year	29.6%	24.2%	31.8%	45.0%
Sharpe ratio (SPY)	0.54 (0.62)	0.37 (0.45)	0.68 (0.81)	0.45 (0.96)
Sortino ratio	0.71	0.48	0.86	0.63
Worst drawdown (SPY)	−58% (−55%)	−58% (−55%)	−52% (−34%)	−35% (−19%)
Beta	0.97	0.86	1.08	1.39
Information ratio	0.16	−0.01	0.25	−0.00
Panel C: Trades				
Trades	710	366	265	62
Win rate	54%	54%	54%	50%
Average win / average loss	+22.4% / −16.3%	+18.1% / −15.5%	+25.4% / −16.9%	+35.8% / −21.6%
Mean / median trade	+4.4% / +2.0%	+2.5% / +1.6%	+6.0% / +2.1%	+7.1% / −0.9%

Note: Each period is a separate portfolio that starts in cash. Intervals resample whole months. Alpha and beta are from a regression of monthly returns on SPY. Annual returns are in Appendix A.

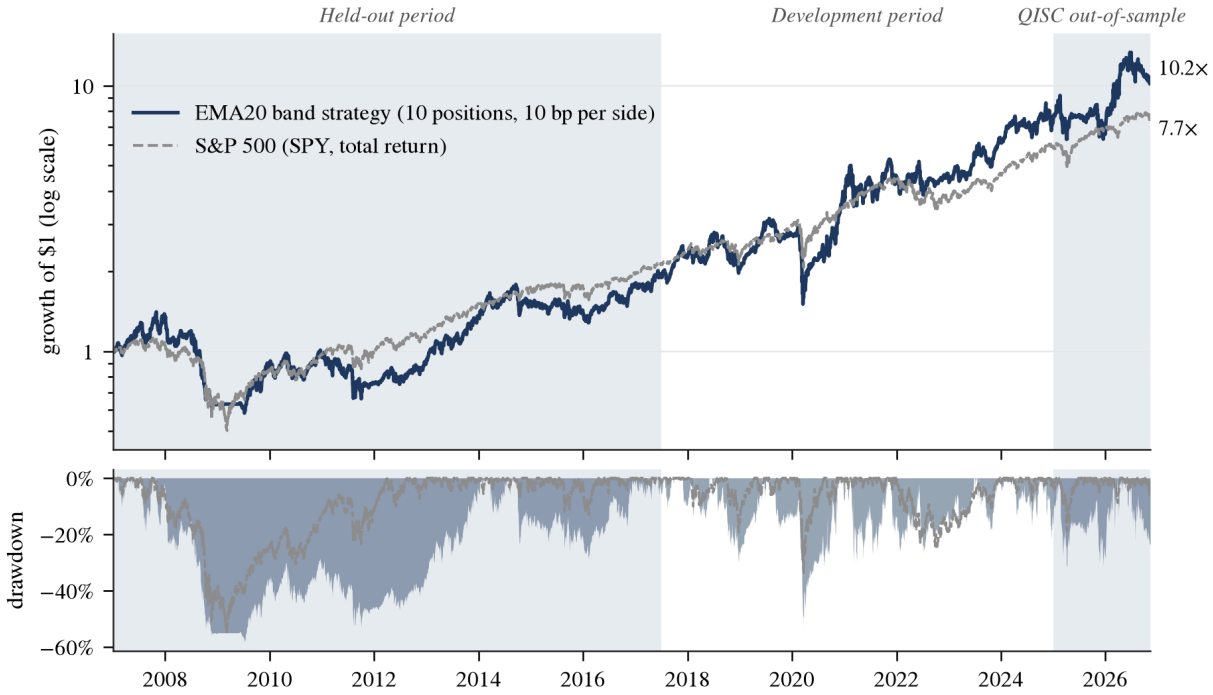


Figure 2. Growth of $1 on a log scale (top) and drawdown from the previous peak (bottom), baseline portfolio and SPY. Shaded regions mark the held-out decade and the QISC out-of-sample window.

The losses are concentrated in abrupt trend reversals: 2008 (−53.0% against −36.8% for SPY), 2011 (−20.2% against +1.9%) and 2025 (−10.7% against +17.7%). Trade returns are strongly skewed (Figure 3): the median trade earns 2.0% and the mean 4.4%, seven of 710 trades more than doubled, and 52 lost at least 30%. The ten best trades account for 44% of summed trade returns.

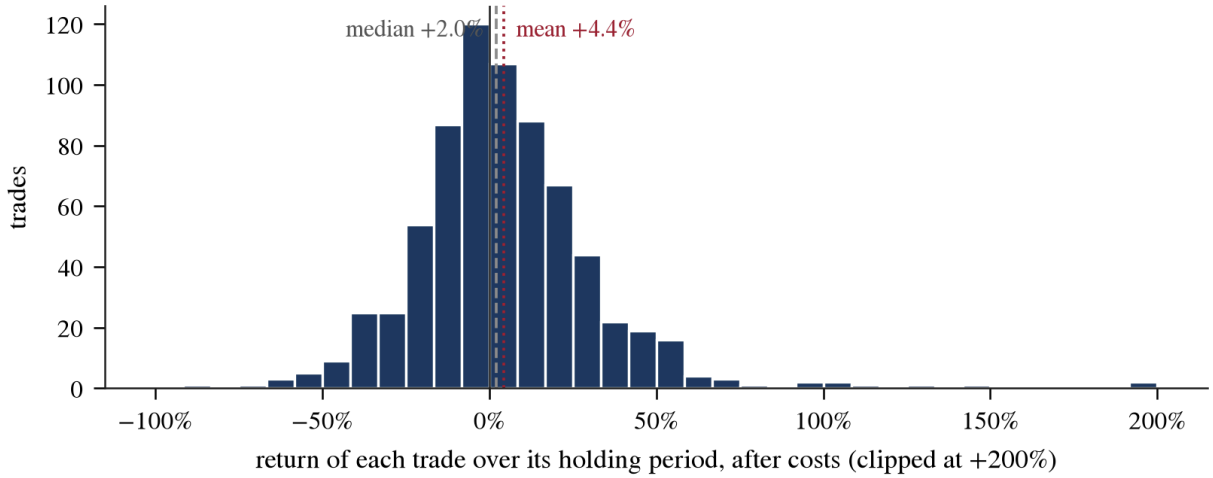


Figure 3. Return of each trade over its holding period, after costs. Returns above +200% are shown at +200%.

5.3. *The timing effect*

Table 5 reports the event study. Over 63 trading days, touch entries earn 5.1%, or 2.6 percentage points above the equal-weighted universe, and no-touch entries into the same top-ranked trends earn 4.2%, or 2.0 points above it. Both intervals exclude zero, so the selection effect (H1) is positive, but it is concentrated in the development period: in the held-out decade neither group beats the universe significantly.

Table 5. Event study: entries into top-ten stocks with and without a band touch.

Horizon	Touch: return	Touch: above universe [95%]	No touch: return	No touch: above universe [95%]	Touch minus no touch, same month [95%]
Panel A: Full sample					
5 days	+0.3%	+0.1% [−0.2%, +0.4%]	−0.0%	−0.2% [−0.4%, +0.1%]	+1.1% [+0.7%, +1.5%]
21 days	+1.5%	+0.9% [+0.2%, +1.6%]	+1.0%	+0.3% [−0.5%, +1.2%]	+1.7% [+1.0%, +2.5%]
63 days	+5.1%	+2.6% [+1.1%, +4.3%]	+4.2%	+2.0% [+0.4%, +3.7%]	+2.3% [+1.0%, +3.5%]
Panel B: By period, 5 days					
Held-out	+0.2%	−0.1% [−0.4%, +0.2%]	−0.1%	−0.3% [−0.6%, −0.0%]	+1.0% [+0.7%, +1.4%]
Development	+0.3%	+0.3% [−0.2%, +0.8%]	+0.0%	−0.1% [−0.5%, +0.3%]	+1.2% [+0.4%, +2.1%]
QISC out-of-sample	+0.5%	+0.5% [−0.8%, +1.8%]	+0.1%	+0.0% [−1.2%, +1.1%]	+0.7% [−0.6%, +2.1%]
Panel C: By period, 63 days					
Held-out	+3.9%	+1.2% [−0.4%, +2.8%]	+2.2%	−0.1% [−1.5%, +1.2%]	+2.1% [+0.2%, +3.9%]
Development	+5.5%	+3.5% [+1.1%, +5.9%]	+4.2%	+2.5% [+0.4%, +4.6%]	+2.6% [+0.8%, +4.5%]
QISC out-of-sample	+11.0%	+8.0% [−2.8%, +20.7%]	+13.8%	+10.8% [−0.9%, +22.8%]	+1.8% [−1.9%, +5.7%]

Note: 4,163 touch and 26,983 no-touch events. Returns run from the next open, net of 10 bp per side. "Above universe" subtracts the equal-weighted return of all eligible stocks over the same days. The last column is the mean over months (226 to 229 in the full sample) of the within-month difference between touch and no-touch returns.

The timing effect (H2) is robust. Within the same month, touch entries beat no-touch entries by 1.1 percentage points over five days, 1.7 over 21 days and 2.3 over 63 days, with every interval excluding zero. The effect is significant in the held-out decade, +1.0 points over five days and +2.1 over 63 days, and has the same sign in the out-of-sample window, where 23 months leave the intervals wide. In pooled averages (Figure 4) the gap widens steadily to 0.9 points at 63 days; it is smaller than the month-matched difference because no-touch entries cluster in strong markets.

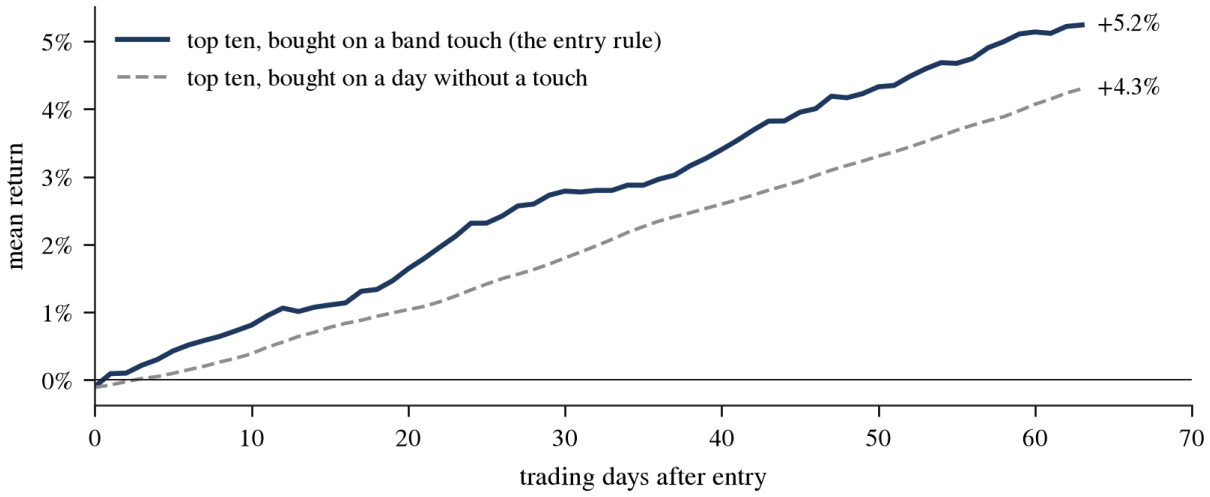


Figure 4. Mean cumulative return from the entry open for top-ten stocks bought on a band touch and without one, pooled over all events.

6. Robustness

6.1. *Exit rules*

A time-based exit is often defended on three grounds: momentum returns accrue slowly, a few large winners carry the strategy, and price-based exits cannot distinguish a pullback from the end of a trend. Table B.1 compares eight alternatives on the same entries. Exits on closes below the band are clearly worse, consistent with that argument. A 40% stop-loss, however, raises the CAGR from 12.4% to 14.8% and the ratio of CAGR to drawdown from 0.21 to 0.29, and an exit on a heavy-volume down day performs almost as well. The evidence on stop-losses in long-only portfolios is mixed: they can reduce risk but often at a cost in return (Lei and Li, 2009; Kaminski and Lo, 2014). We do not adopt either alternative, because both were selected after seeing the results and would require their own out-of-sample test.

6.2. *Position count, holding period and costs*

Figure 5 and Table B.2 vary one parameter at a time. With three positions the portfolio loses 87% at its worst; with five, 71%. Additional positions mainly reduce the drawdown, and twenty give the best ratio of CAGR to drawdown (0.28), but the ranking of portfolio sizes is unstable across periods. The holding period matters little between 42 and 126 days, while 21 days is clearly worse. Costs matter little: the portfolio trades about 7.2 times its capital a year, counting purchases and sales, so each additional basis point per side costs about 0.07 percentage points a year, and even 50 bp leaves 9.4%.

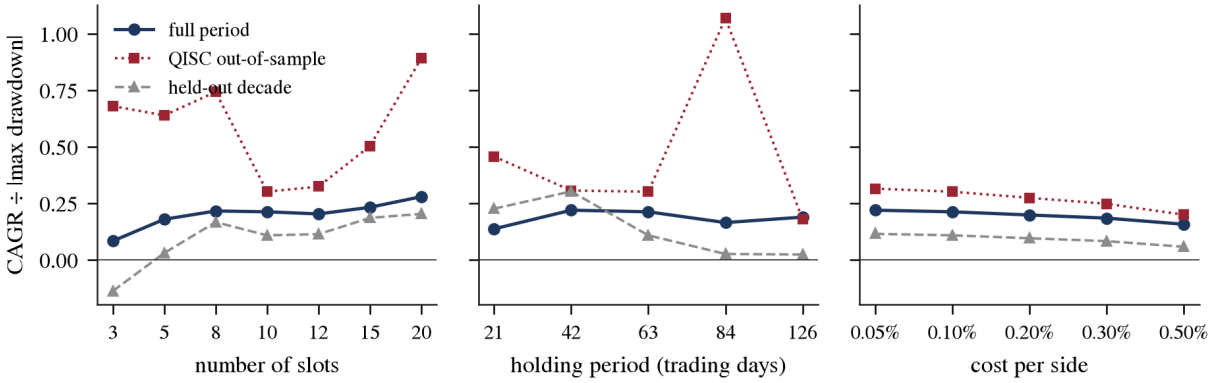


Figure 5. CAGR divided by the absolute maximum drawdown as one parameter of the baseline portfolio varies, for the full sample, the held-out decade and the QISC out-of-sample window.

6.3. *Entry parameters*

Table B.3 varies the touch window, the tolerance above the band and the EMA span, with the trend screen unchanged. The baseline setting is not the best cell: a ten-day window with a close strictly inside the band raises the CAGR to 15.9%. Neighbouring cells, however, show no smooth pattern, and the best tolerance changes from one window to the next, which points to noise rather than a better parameter. Spans from 10 to 40 days give CAGRs between 8.0% and 14.5%, with the shortest span clearly the worst.

6.4. *Stress tests*

Table 6 reports the stress tests described in the casebook and four further checks. Delaying every trade by one day does not hurt (13.0% a year). Removing the five largest contributors lowers the return to 11.7% and the monthly excess to 0.22%. Buying the same stocks without waiting for a touch earns 11.9% over the full sample but only 0.7% a year in the out-of-sample window, against 10.7% for the baseline: the timing rule is what separates the two. Counting pullbacks only once confirmed lowers the return to 10.4%, almost entirely in the development period, and leaves every conclusion unchanged. Starting the portfolio in different months moves the held-out CAGR between 3.3% and 7.6% and the development CAGR between 12.0% and 18.7%.

Table 6. Stress tests, January 2007 to November 2026.

Test	Per year	Worst drawdown	Sharpe	Monthly excess [95%]
Baseline portfolio	+12.4%	−58%	0.54	+0.29% [−0.5%, +1.1%]
Execution one day later	+13.0%	−56%	0.56	+0.32% [−0.5%, +1.1%]
Without the five largest contributors	+11.7%	−58%	0.53	+0.22% [−0.5%, +1.0%]
Bought without a touch	+11.9%	−58%	0.51	+0.32% [−0.6%, +1.2%]
Pullbacks counted only once confirmed	+10.4%	−58%	0.48	+0.15% [−0.7%, +0.9%]
New positions only while SPY is above its 200-day average	+13.2%	−54%	0.59	+0.32% [−0.5%, +1.1%]
Twelve start months, held-out decade	+3.3% to +7.6%	−52% to −60%		
Twelve start months, development period	+12.0% to +18.7%	−50% to −56%		

Note: The market filter is exploratory: it was specified after the 2008 drawdown had been observed and is counted as a trial in Section 6.5.

The market filter is the most tempting fix. Opening positions only while SPY is above its 200-day average, which held on 79% of days, cuts the worst drawdown from 58% to 54% and raises the Sharpe ratio to 0.59, still below SPY's, without creating a significant excess return.

6.5. *Multiple testing*

Section 6 reports 45 variants, counting the baseline each time it appears. If none had any true edge, the expected maximum annualised Sharpe ratio across 45 trials with the observed dispersion would be 0.17. The deflated Sharpe ratio of the baseline, the probability that its true Sharpe ratio exceeds that benchmark after allowing for skewness and fat tails, is 0.951, just above the conventional 0.95. It reflects mostly the portfolio's equity exposure; its advantage over SPY is not significant (Table 4).

7. Implementation and risk

Capacity. At entry, the median stock bought trades $31.1 million a day (60-day median) and the 5th percentile $2.7 million. Keeping each position below 5% of that value, as the casebook

requires, limits a position to about $130,000 in the least liquid names and $1.6 million in the median name. Positions are sized on a compounding book that starts at $1 million: 97% of entries meet the 5% limit while the book is below $2 million, but only about 70% in 2023–2025, when it has grown to several million. A larger book would have to raise the liquidity floor as it grows, which would change the universe.

Execution. All trades are market-on-open orders placed after the previous close, so the strategy needs one decision a day. Because every trade happens at the open, returns combine overnight and intraday components, which respond differently to news (Bahcivan, 2025); at a holding period of 63 days, the opening spread in small stocks, not commissions, is the binding cost.

Risk. Beta is 0.97 over the full sample and 1.39 in the out-of-sample window, and volatility is about one and a half times that of SPY. Investors should expect a drawdown of 50% to 60% and about one trade in fourteen to lose 30% or more. Losses cluster in abrupt market reversals, the momentum-crash pattern of Daniel and Moskowitz (2016); volatility scaling, which helps conventional momentum portfolios (Barroso and Santa-Clara, 2015), is a natural next test.

Kill criteria. We would stop trading if the drawdown exceeded 65%, beyond any backtest path, or if the month-matched advantage of touch over no-touch entries turned negative over a trailing 24-month window.

8. Conclusion

A concentrated version of this strategy earns 39.1% a year in the sample on which it was developed. Over twenty years and under the QISC constraints, the portfolio earns 12.4% a year against 10.8% for the S&P 500, with more risk and no significant excess return; it beats the index only in the development period. Its design choices, the five-position portfolio and the time-only exit among them, do not carry over to the held-out decade.

The pullback entry does. Waiting for a stock in a strong trend to dip into its 20-day EMA band and close back above it improves the entry by about one percentage point over the following week and two points over three months, relative to buying a comparable stock in the same month without the pullback, before as well as after the development period. We therefore recommend the band touch as an entry filter to combine with other selection rules, rather than the portfolio in its current form, and would test every further refinement, including the exits and market filter in Section 6, on data not used to choose it.

Statement of Originality and AI Disclosure

We declare that this paper and the accompanying code are the original work of Team [Team Name], that all sources are cited, and that the submission has not been submitted to any other competition. Generative AI was used to assist with programming the data pipeline and with formatting tables and figures. The strategy, the analysis and the written paper are the team's own work, and the team takes full responsibility for the content.

Code and data availability

The repository `qisc-[team-name]` reproduces every table and figure with one command, `python run.py`. It downloads the symbol directory, the Nasdaq screener and the prices, computes the point-in-time scores, runs every test, and writes the tables, the figures, the trade list and `output/positions.csv` in the format required by the competition. All data are publicly available; a full run takes about 30 minutes, most of it the price download.

References

Bahcivan, H., 2025. Day and night expected returns under overnight information shocks: new tug-of-war pattern. *Finance Research Letters* 86, 108591.

Bailey, D.H., López de Prado, M., 2014. The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management* 40 (5), 94–107.

Bajgrowicz, P., Scaillet, O., 2012. Technical trading revisited: false discoveries, persistence tests, and transaction costs. *Journal of Financial Economics* 106 (3), 473–491.

Barroso, P., Santa-Clara, P., 2015. Momentum has its moments. *Journal of Financial Economics* 116 (1), 111–120.

Brock, W., Lakonishok, J., LeBaron, B., 1992. Simple technical trading rules and the stochastic properties of stock returns. *Journal of Finance* 47 (5), 1731–1764.

Daniel, K., Moskowitz, T.J., 2016. Momentum crashes. *Journal of Financial Economics* 122 (2), 221–247.

Frazzini, A., 2006. The disposition effect and underreaction to news. *Journal of Finance* 61 (4), 2017–2046.

Han, Y., Yang, K., Zhou, G., 2013. A new anomaly: the cross-sectional profitability of technical analysis. *Journal of Financial and Quantitative Analysis* 48 (5), 1433–1461.

Harvey, C.R., Liu, Y., Zhu, H., 2016. … and the cross-section of expected returns. *Review of Financial Studies* 29 (1), 5–68.

Jegadeesh, N., 1990. Evidence of predictable behavior of security returns. *Journal of Finance* 45 (3), 881–898.

Jegadeesh, N., Titman, S., 1993. Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance* 48 (1), 65–91.

Jegadeesh, N., Titman, S., 2001. Profitability of momentum strategies: an evaluation of alternative explanations. *Journal of Finance* 56 (2), 699–720.

Kaminski, K.M., Lo, A.W., 2014. When do stop-loss rules stop losses? *Journal of Financial Markets* 18, 234–254.

Lehmann, B.N., 1990. Fads, martingales, and market efficiency. *Quarterly Journal of Economics* 105 (1), 1–28.

Lei, A.Y.T., Li, H., 2009. The value of stop loss strategies. *Financial Services Review* 18 (1), 23–51.

Marshall, B.R., Cahan, R.H., Cahan, J.M., 2008. Does intraday technical analysis in the U.S. equity market have value? *Journal of Empirical Finance* 15, 199–210.

McLean, R.D., Pontiff, J., 2016. Does academic research destroy stock return predictability? *Journal of Finance* 71 (1), 5–32.

Moskowitz, T.J., Ooi, Y.H., Pedersen, L.H., 2012. Time series momentum. *Journal of Financial Economics* 104 (2), 228–250.

Nagel, S., 2012. Evaporating liquidity. *Review of Financial Studies* 25 (7), 2005–2039.

Patel, J., Shah, S., Thakkar, P., Kotecha, K., 2015. Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications* 42, 259–268.

Sezer, O.B., Ozbayoglu, A.M., 2018. Algorithmic financial trading with deep convolutional neural networks: time series to image conversion approach. *Applied Soft Computing* 70, 525–538.

Shefrin, H., Statman, M., 1985. The disposition to sell winners too early and ride losers too long: theory and evidence. *Journal of Finance* 40 (3), 777–790.

Sullivan, R., Timmermann, A., White, H., 1999. Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance* 54 (5), 1647–1691.

Tsinaslanidis, P., Guijarro, F., Voukelatos, N., 2022. Automatic identification and evaluation of Fibonacci retracements: empirical evidence from three equity markets. *Expert Systems with Applications* 187, 115893.

Appendix A. Annual returns

Table A.1. Calendar-year returns of the baseline portfolio.

Year	Portfolio	SPY	Difference
2007	+34.6%	+5.3%	+29.3%
2008	−53.0%	−36.8%	−16.3%
2009	+39.7%	+26.4%	+13.3%
2010	+7.4%	+15.1%	−7.6%
2011	−20.2%	+1.9%	−22.1%
2012	+20.5%	+16.0%	+4.5%
2013	+46.3%	+32.3%	+14.0%
2014	+13.4%	+13.5%	−0.1%
2015	−7.3%	+1.2%	−8.6%
2016	+24.4%	+12.0%	+12.4%
2017	+28.7%	+21.7%	+7.0%
2018	−7.5%	−4.6%	−3.0%
2019	+33.2%	+31.2%	+2.0%
2020	+33.0%	+18.3%	+14.6%
2021	+24.2%	+28.7%	−4.6%
2022	−0.6%	−18.2%	+17.6%
2023	+38.1%	+26.2%	+11.9%
2024	+17.2%	+24.9%	−7.7%
2025	−10.7%	+17.7%	−28.4%
2026*	+55.2%	+11.7%	+43.5%

Note: The portfolio runs continuously from January 2007 and beats SPY in eleven of twenty years. * 2026 to 11 November.

Appendix B. Robustness tables

Table B.1. Exit rules, January 2007 to November 2026.

Exit rule (entries unchanged)	Trades	Win rate	Per year	Worst drawdown	CAGR ÷ drawdown	Sharpe
Hold 63 days (baseline)	710	54%	+12.4%	−58%	0.21	0.54
Close below the band	2,972	32%	+3.6%	−46%	0.08	0.27
Three closes below the band	1,942	35%	+6.5%	−56%	0.12	0.38
Five closes below the band	1,452	37%	+8.7%	−45%	0.19	0.45
Heavy-volume down day (≥ 3×)	995	52%	+14.7%	−52%	0.28	0.62
Heavy-volume down day (≥ 5×)	809	52%	+14.3%	−54%	0.26	0.61
Three closes below the band or volume ≥ 3×	2,122	36%	+5.9%	−56%	0.11	0.36
Stop at −25%	782	52%	+10.9%	−47%	0.23	0.51
Stop at −40%	721	54%	+14.8%	−51%	0.29	0.63

Note: Exits other than the clock are decided on a close and executed at the next open; every position is sold after 63 days at the latest. Heavy-volume down day: a lower close on volume of at least the stated multiple of its 50-day median.

Table B.2. Position count, holding period and costs.

Variant	Trades	Per year	Worst drawdown	Sharpe	CAGR ÷ drawdown: full	Held-out	Development	QISC OOS
Panel A: Number of positions								
3	217	+7.2%	−87%	0.38	0.08	−0.14	0.59	0.68
5	360	+12.8%	−71%	0.52	0.18	0.03	0.31	0.64
8	569	+13.3%	−62%	0.56	0.22	0.17	0.29	0.74
10 (baseline)	710	+12.4%	−58%	0.54	0.21	0.11	0.34	0.30
12	846	+11.3%	−56%	0.52	0.20	0.11	0.35	0.32
15	1,033	+11.4%	−49%	0.54	0.23	0.19	0.22	0.50
20	1,348	+12.6%	−45%	0.60	0.28	0.20	0.26	0.89
Panel B: Holding period (trading days)								
21	1,737	+6.9%	−51%	0.39	0.14	0.23	0.06	0.46
42	1,017	+11.8%	−54%	0.53	0.22	0.30	0.23	0.31
63 (baseline)	710	+12.4%	−58%	0.54	0.21	0.11	0.34	0.30
84	544	+10.3%	−63%	0.48	0.16	0.03	0.32	1.07
126	373	+13.1%	−69%	0.55	0.19	0.02	0.38	0.18
Panel C: Cost per side								
5 bp	710	+12.8%	−58%	0.56	0.22	0.11	0.35	0.31
10 bp (baseline)	710	+12.4%	−58%	0.54	0.21	0.11	0.34	0.30
20 bp	710	+11.6%	−59%	0.52	0.20	0.10	0.32	0.27
30 bp	710	+10.9%	−59%	0.50	0.18	0.08	0.31	0.25
50 bp	710	+9.4%	−60%	0.45	0.16	0.06	0.27	0.20

Note: Each row changes one parameter of the baseline. The last three columns are separate portfolios per period.

Table B.3. Entry parameters.

Touch window or EMA span	Tolerance	Trades	Per year	Worst drawdown	CAGR ÷ drawdown	Sharpe
Panel A: Touch window						
3 days	0%	650	+11.9%	−50%	0.24	0.54
3 days	1%	692	+11.8%	−54%	0.22	0.53
3 days	2%	715	+11.6%	−55%	0.21	0.52
5 days	0%	673	+13.4%	−56%	0.24	0.58
5 days (baseline)	1%	710	+12.4%	−58%	0.21	0.54
5 days	2%	724	+13.7%	−58%	0.24	0.58
10 days	0%	701	+15.9%	−49%	0.33	0.65
10 days	1%	721	+11.5%	−54%	0.21	0.51
10 days	2%	729	+12.4%	−49%	0.25	0.54
Panel B: EMA span of the band						
10 days	1%	730	+8.0%	−58%	0.14	0.41
15 days	1%	723	+10.8%	−54%	0.20	0.49
20 days (baseline)	1%	710	+12.4%	−58%	0.21	0.54
30 days	1%	655	+14.5%	−53%	0.27	0.62
40 days	1%	570	+11.5%	−47%	0.24	0.56

Note: Panel B keeps the five-day window. The trend screen is unchanged throughout.

Appendix C. Trades

The full list of 710 trades is in `output/trades.csv`. Prices are split- and dividend-adjusted opens; returns are after 10 bp per side.

Table C.1. The ten best trades.

#	Stock	Bought	Buy price	Sold	Sell price	Return
1	AXTI	2026-01-22	19.95	2026-04-23	82.56	+313.6%
2	PACB	2020-11-03	11.77	2021-02-04	34.87	+196.1%
3	CRDF	2020-08-05	6.49	2020-11-03	15.83	+143.7%
4	LITE	2026-01-08	394.3	2026-04-10	908.09	+130.1%
5	ZM	2020-07-21	268.75	2020-10-19	572.5	+112.8%
6	MLTX	2023-06-16	27.47	2023-09-18	55.69	+102.5%
7	ENPH	2019-04-29	10.38	2019-07-29	21.0	+102.1%
8	CELH	2020-09-23	7.0	2020-12-22	13.83	+97.4%
9	MOS	2007-10-04	35.46	2008-01-04	68.55	+93.1%
10	UCTT	2017-03-09	13.81	2017-06-08	24.9	+80.1%

Table C.2. The ten worst trades.

#	Stock	Bought	Buy price	Sold	Sell price	Return
1	NNBR	2008-09-15	13.53	2008-12-12	1.79	−87.0%
2	AIRS	2025-11-07	7.3	2026-02-10	2.25	−69.4%
3	CRMD	2015-04-22	45.05	2015-07-22	16.55	−63.5%
4	HP	2008-07-15	36.79	2008-10-13	14.21	−61.6%
5	FTK	2007-10-25	290.58	2008-01-28	121.26	−58.5%
6	CDE	2019-12-30	8.01	2020-03-31	3.38	−58.1%
7	CROX	2007-10-22	64.65	2008-01-23	28.82	−55.6%
8	VSEC	2007-10-29	25.03	2008-01-30	11.83	−53.0%
9	SGI	2020-01-16	20.91	2020-04-17	10.15	−51.7%
10	ASTE	2007-10-15	49.95	2008-01-15	24.7	−50.7%

Table C.3. Every trade entered in the QISC out-of-sample window, January 2025 to November 2026.

#	Stock	Bought	Buy price	Sold	Sell price	Return
1	ETON	2025-01-14	14.14	2025-04-15	13.7	−3.3%
2	KINS	2025-01-16	14.86	2025-04-17	17.17	+15.4%
3	GHM	2025-01-16	43.97	2025-04-17	30.1	−31.7%
4	PLTR	2025-01-21	73.24	2025-04-22	93.01	+26.8%
5	RKLB	2025-01-23	29.0	2025-04-24	20.31	−30.2%
6	WGS	2025-01-27	76.51	2025-04-28	111.39	+45.4%
7	UAL	2025-01-28	105.83	2025-04-29	68.46	−35.5%
8	AHR	2025-03-17	28.25	2025-06-16	34.5	+21.9%
9	FSCO	2025-03-18	5.67	2025-06-17	6.02	+5.9%
10	GCMG	2025-03-19	12.47	2025-06-18	11.07	−11.4%
11	CME	2025-04-15	251.89	2025-07-17	265.26	+5.1%
12	ORI	2025-04-17	34.15	2025-07-21	33.49	−2.1%
13	CNP	2025-04-22	35.75	2025-07-23	36.61	+2.2%
14	VRSN	2025-04-24	248.27	2025-07-25	290.53	+16.8%
15	KINS	2025-04-25	18.15	2025-07-28	15.68	−13.8%
16	CAH	2025-04-28	135.85	2025-07-29	155.05	+13.9%

#	Stock	Bought	Buy price	Sold	Sell price	Return
17	TXNM	2025-05-05	51.07	2025-08-05	54.57	+6.6%
18	AHR	2025-06-24	35.45	2025-09-23	40.95	+15.3%
19	FSCO	2025-06-25	6.05	2025-09-24	6.22	+2.6%
20	TDUP	2025-07-07	7.96	2025-10-03	9.3	+16.6%
21	AEVA	2025-07-18	31.0	2025-10-16	19.7	−36.7%
22	DASH	2025-07-21	240.44	2025-10-17	263.86	+9.5%
23	PARR	2025-07-25	31.85	2025-10-23	37.89	+18.8%
24	VRT	2025-07-29	142.89	2025-10-27	189.79	+32.6%
25	CRDO	2025-07-29	110.0	2025-10-27	162.32	+47.4%
26	PAYS	2025-07-30	7.76	2025-10-28	5.81	−25.3%
27	AKBA	2025-08-07	3.2	2025-11-05	2.08	−35.3%
28	AHR	2025-10-01	41.14	2025-12-31	47.01	+14.1%
29	AMLX	2025-10-08	13.68	2026-01-08	12.17	−11.2%
30	AMPX	2025-10-21	13.16	2026-01-22	11.71	−11.2%
31	KTOS	2025-10-22	90.0	2026-01-23	113.83	+26.3%
32	REZI	2025-10-23	29.03	2026-01-26	24.99	−14.1%
33	BE	2025-10-24	106.0	2026-01-27	141.0	+32.8%
34	LEU	2025-10-27	391.15	2026-01-28	313.0	−20.2%
35	HUT	2025-10-27	48.65	2026-01-28	63.74	+30.8%
36	JMIA	2025-10-28	11.28	2026-01-29	12.87	+13.9%
37	AIRS	2025-11-07	7.3	2026-02-10	2.25	−69.4%
38	NRGV	2026-01-05	5.04	2026-04-07	3.09	−38.9%
39	LITE	2026-01-08	394.3	2026-04-10	908.09	+130.1%
40	AXTI	2026-01-22	19.95	2026-04-23	82.56	+313.6%
41	ARWR	2026-01-23	69.25	2026-04-24	74.29	+7.1%
42	HYMC	2026-02-04	43.46	2026-05-06	37.58	−13.7%
43	ARIS	2026-02-04	20.23	2026-05-06	18.38	−9.3%
44	LASR	2026-02-06	47.59	2026-05-08	76.96	+61.5%
45	AHR	2026-02-06	47.78	2026-05-08	49.03	+2.4%
46	TTI	2026-02-09	11.74	2026-05-11	9.86	−16.2%
47	VSAT	2026-02-10	45.67	2026-05-12	71.73	+56.9%
48	BW	2026-04-07	16.24	2026-07-08	10.27	−37.0%
49	LITE	2026-04-30	895.17	2026-07-31	748.0	−16.6%
50	CIEN	2026-05-01	527.25	2026-08-03	365.0	−31.0%
51	ORKA	2026-05-05	67.17	2026-08-05	98.72	+46.8%
52	SPHR	2026-05-19	136.73	2026-08-19	165.84	+21.1%
53	INTT	2026-05-20	17.87	2026-08-20	12.62	−29.6%
54	VICR	2026-05-21	262.96	2026-08-21	211.06	−19.9%
55	SYRE	2026-05-21	70.89	2026-08-21	108.85	+53.3%
56	LASR	2026-05-22	77.36	2026-08-24	46.67	−39.9%
57	VIAV	2026-05-26	51.0	2026-08-25	38.26	−25.2%
58	HSHP	2026-07-08	14.22	2026-10-06	17.58	+23.4%
59	DFTX	2026-07-31	44.98	2026-10-29	38.32	−15.0%
60	VSAT	2026-08-03	75.41	2026-10-30	74.26	−1.7%
61	CLYM	2026-08-05	12.56	2026-11-03	9.86	−21.7%